Lab 02

COMP1323 Networks and Security

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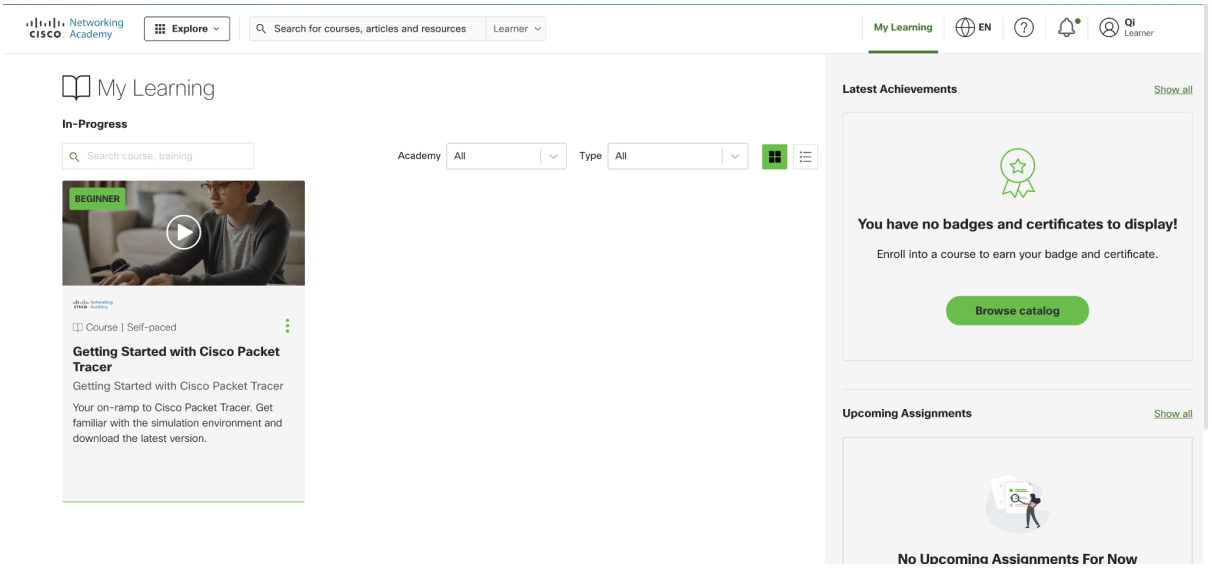
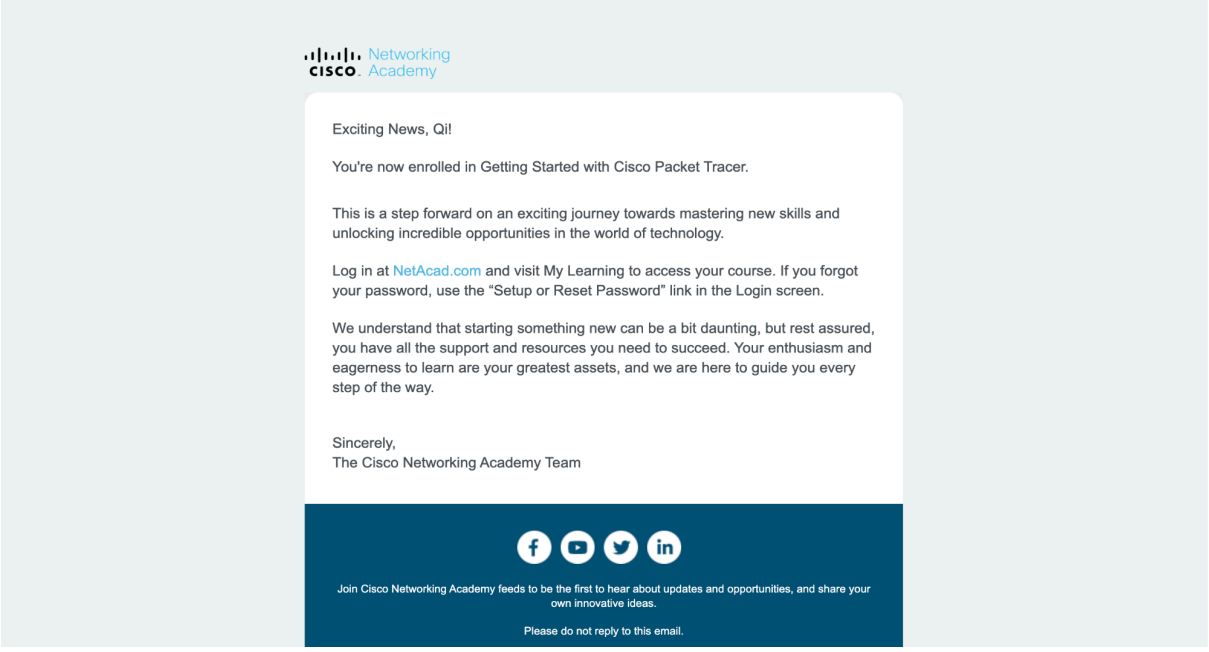
Activity 1:

• Screenshot of course enrollment



1



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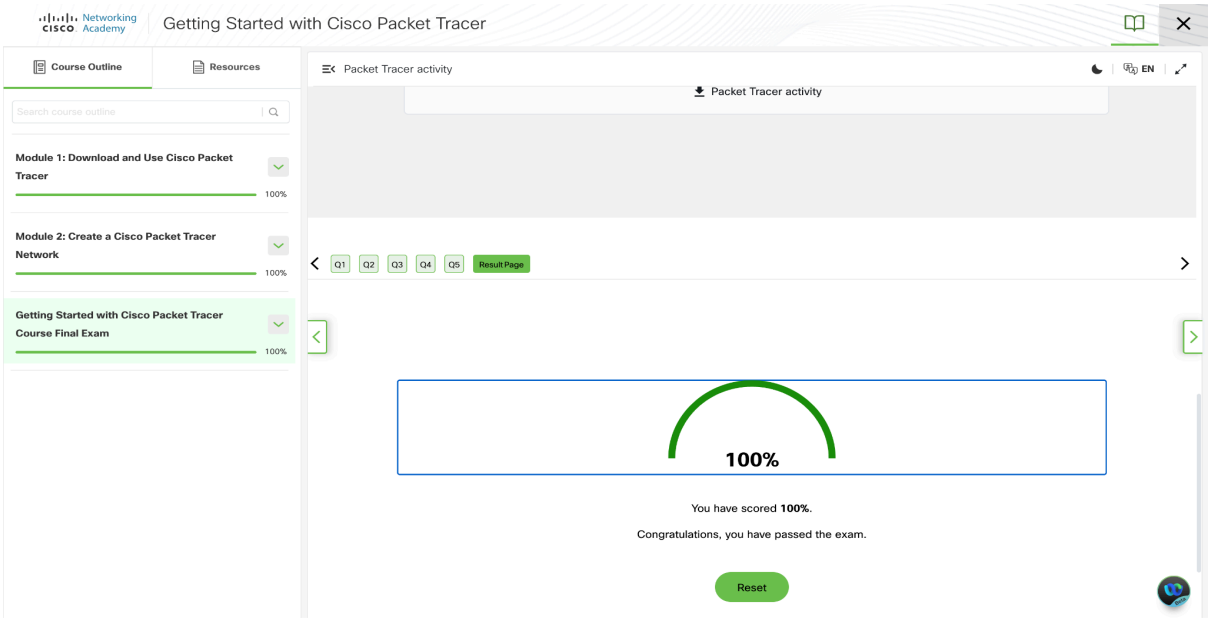
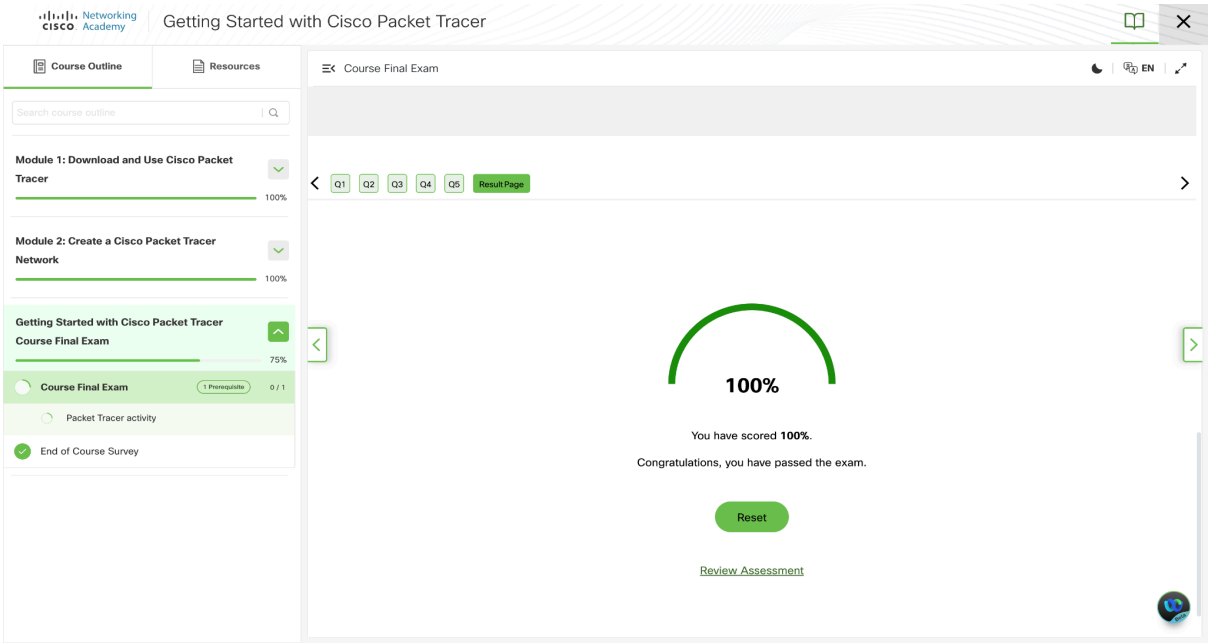
[1] First screenshot of course enrollment
[2] Second screenshot of course enrollment

- Screenshot showing course progress



[3]

- Screenshot of final assessment completion



[4]

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[3] Screenshot showing course progress
[4] Screenshot of final assessment completion

Activity 2:

Official certificate:

Networking Academy

This certificate is awarded to

Qi Liu

for successfully completing

Getting Started with Cisco Packet Tracer

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer
Director
Cisco Networking Academy

03 Mar 2026
Completion Date

[5]

Activity 3:

1. What is the purpose of Cisco Packet Tracer?
Cisco Packet Tracer is a program used to practice building and testing computer networks.

2. What are the two main operating modes in Packet Tracer?
RealTime Mode and Simulation Mode.

3. Explain the difference between Realtime Mode and Simulation Mode.
Realtime Mode: The network works normally and runs immediately in real time.
Simulation Mode: Everyone can see how data move step by step with a simulated GUI.

[5] Official certificate

4. What is the function of the Logical Workspace?
It is where we use diagrams to build and connect network devices together.

5. What is the Physical Workspace used for?
It shows how devices are placed in real workspaces like rooms and buildings, the same as what we can see in our real world .

6. List at least four types of network devices available in Packet Tracer.
Router, Switch, Server, Wireless Access Point, Laptop, PC, Printer, File Server, and Smartphone .

7. What is the role of a switch in a network?
It can be used to connect multiple devices (as I have already explained in the last question 6) to a single local area network (LAN) with the help of Ethernet cables. It can receive the data and then send it to the particular port to which the device to be connected is connected according to the destination MAC addresses. It belongs to Network Layer.

8. What is the function of a router?
A router is a networking device, also known as a Network Layer device, which connects two or more networks to enable data exchange. The device directs traffic flow by reading the data packets and their destination IP addresses, then routing them appropriately using routing tables.

9. What is an IP address and why is it important?
An IP address is a unique numerical label given to each and every device on a network, such as 192.168.0.1, which allows them to communicate with each other. It is a crucial term as it is a virtual identifier that helps information reach its correct destination

10. What happens if two devices are in different networks without a router?
They cannot communicate with each other because they are on different subnets so that they will send traffic to a non-existent default gateway.

11. What is the purpose of using cables in Packet Tracer?
Cables are used for connecting devices so they can send and receive data.

12. What is the difference between straight-through and crossover cables?
Straight-through: Connecting different devices (Laptop to Switch, Printer to Switch).
Crossover: Connecting the same devices (PC to PC, Switch to another Switch).

13. What does a successful ping test indicate?
It means the devices can communicate with each other. A ping test is frequently used to test the error device when the network topology goes wrong.

14. Why is Simulation Mode useful for networking students?
Because everyone can see how data move step by step. And it is much easier to see where the data stop if there is something wrong with the network.

15. How does Packet Tracer help in understanding real-world networking concepts?
It allows me to practice building networks in a virtual and graphical environment. And it also helps me to identify each kind of devices which would be used in a real-world networking.